MATH 2355 - MIDTERM EXAM
INSTRUCTOR: ALISTAIR SAVAGE
OCTOBER 20, 2008

Surname _____ First Name _____

Student # _____

Instructions:

(a) This test consists of 4 questions for a total of 27 points.
(b) All work to be considered for grading must be written in the space provided. The reverse side of pages is for scrap work only and any material written there will **NOT** be considered for marks (unless you clearly indicate that your answer is continued on the reverse of a page).
(c) You must show/justify your work to obtain full marks.
(d) Write your student number at the top of each page in the space provided.
(e) No notes, book, calculators or scrap paper are allowed.
(f) Good luck!

Please do not write in the table below.

Question	1	2	3	4	Total
Maximum	6	9	4	8	27
Score					

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1. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $f(x, y) = (3 - y, 2 - x)$.
   (a) **(2 points)** Show that f is an isometry.

   (b) **(2 points)** If f orientation preserving or orientation reversing?

   (c) **(2 points)** Is f a reflection? If so, find the line of reflection. If not, give reasons why it is not.

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2. In each of the following, identify the isometry as either
   (i) a reflection R_L in a line L (give an equation for L),
   (ii) a translation by a vector v (find v),
   (iii) a rotation about a point $c \in \mathbb{R}^2$ (find c), or
   (iv) a glide reflection associated with a line L and a translation v parallel to L (find L and v).

As usual, you must justify your answer.

(a) **(3 points)** $f : \mathbb{R}^2 \to \mathbb{R}^2$ given by $f(x) = Ax + b$, $x \in \mathbb{R}^2$, where

$$A = \begin{bmatrix} 4/5 & -3/5 \\ 3/5 & 4/5 \end{bmatrix}, \quad b = \begin{bmatrix} 3/5 \\ 1/5 \end{bmatrix}.$$

(b) **(3 points)** $g : \mathbb{R}^2 \to \mathbb{R}^2$ given by $g(x) = Bx$, $x \in \mathbb{R}^2$, where

$$B = \begin{bmatrix} -4/5 & -3/5 \\ -3/5 & 4/5 \end{bmatrix}.$$

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(c) **(3 points)** $h : \mathbb{R}^2 \to \mathbb{R}^2$ given by $h(x) = Bx + c$, $x \in \mathbb{R}^2$, where B is the matrix from the previous part and

$$c = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

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3. **(4 points)** Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be an isometry. We know from class that f can be expressed as

$$f(x) = Ax + b, \quad x \in \mathbb{R}^n,$$

for an $n \times n$ orthogonal matrix A and a vector $b \in \mathbb{R}^n$. Prove that such an expression is *unique*. That is, show that if f is expressed as $f(x) = Cx + d$ for an $n \times n$ orthogonal matrix C and $d \in \mathbb{R}^n$, then we must have $A = C$ and $b = d$.

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4.

(a) **(2 points)** Show that if $f, g : \mathbb{R}^n \to \mathbb{R}^n$ are similarity transformations with scaling factors k_f and k_g respectively, then the composition $f \circ g$ is also a similarity transformation. What is its scaling factor?

(b) **(2 points)** Show that any similarity transformation is one-to-one. That is, show that if $f : \mathbb{R}^n \to \mathbb{R}^n$ is a similarity transformation and $f(x) = f(y)$ for some $x, y \in \mathbb{R}^n$, then $x = y$.

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(c) **(2 points)** Show that any similarity transformation is onto. That is, show that if $f : \mathbb{R}^n \to \mathbb{R}^n$ is a similarity transformation and $y \in \mathbb{R}^n$, then there is some $x \in \mathbb{R}^n$ such that $f(x) = y$.

(d) **(2 points)** The previous two parts show that any similarity transformation has an inverse (even if you were not able to show this, you may assume it for the purposes of the current part). If $f : \mathbb{R}^n \to \mathbb{R}^n$ is a similarity transformation with scaling factor k, show that f^{-1} is also a similarity transformation. What is its scaling factor?